

COVID-19

Key facts

- COVID-19 is a disease caused by the SARS-CoV-2 coronavirus. There are a range of signs and symptoms, but the most common symptoms reported for currently-circulating variants are fever, chills, and sore throat.
 - Most people make a full recovery without needing treatment. People with severe disease or at risk of developing severe disease should seek medical care as soon as possible.
 - Nearly 780 million cases and more than 7.1 million deaths due to COVID-19 have been reported worldwide since December 2019, but the actual number is thought to be higher.
 - Over 13.64 billion COVID-19 vaccine doses had been administered by the end of 2024. WHO continues to recommend COVID-19 vaccination using a risk-based approach. WHO recommends a single dose for persons who have not yet received a COVID-19 vaccine. For those at high risk of severe COVID-19, revaccination may be needed 6-12 months after the most recent dose.
 - COVID-19 continues to be a significant public health concern, with global circulation of the SARS-CoV-2 virus and hospitalizations and deaths continuing to occur.
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Overview

COVID-19 is caused by the SARS-CoV-2 virus, which spreads through the air in infectious respiratory particles from an infected person. Transmission is more likely in close contact or shared indoor spaces. The virus can also spread by touching contaminated surfaces and then the eyes, nose or mouth.

Because of increasing immunity from past infections and/or vaccination, most people recover without treatment. About 6% develop post COVID-19 condition (PCC, or long COVID), however, which can cause lasting symptoms such as memory issues, fatigue or breathing difficulties. PCC can affect daily life and requires individualized care.

Older adults, people with underlying health conditions, and unvaccinated individuals are at higher risk of severe illness. Conditions that increase risk include diabetes, heart or lung disease, obesity and weakened immune systems.

Health and care workers face higher exposure risk due to their work environment.

Vaccination remains a key preventive measure against severe disease. COVID-19 vaccines continue to provide strong protection against severe illness and death.

Symptoms

People may experience different symptoms with COVID-19.

The most common symptoms are:

- fever
- chills
- sore throat.

Less common symptoms are:

- muscle aches and heavy arms or legs
- severe fatigue or tiredness
- runny or blocked nose, or sneezing
- headache
- sore eyes

- dizziness
- new and persistent cough
- tight chest or chest pain
- shortness of breath
- hoarse voice
- numbness or tingling
- appetite loss, nausea, vomiting, abdominal pain or diarrhoea
- loss or change of sense of taste or smell
- difficulty sleeping.

Immediate medical attention should be sought by people at risk of developing severe disease or with the following severe symptoms:

- difficulty breathing, especially at rest, or unable to speak in sentences
- confusion
- drowsiness or loss of consciousness
- persistent pain or pressure in the chest
- skin being cold or clammy, or turning pale or a bluish colour
- loss of speech or movement.

People who have underlying health conditions are at higher risk of developing severe disease when they have COVID-19; they should seek medical help early if worried about their condition. These include:

- people with high blood pressure, diabetes, obesity, chronic lung, heart, liver and/or kidney diseases, cancer or dementia;
- people with rheumatological problems;
- women who are pregnant; and
- people with immunosuppression, such as those taking immunosuppressive medication or living with HIV.

The potential consequences of severe COVID-19 include respiratory failure, sepsis, thromboembolism (blood clots), and multiorgan failure, including injury of the heart, liver or kidneys, and death.

In rare situations, children can develop a severe inflammatory syndrome a few weeks after infection and may require medical attention and hospitalization.

Some people who have had COVID-19, whether they needed hospitalization or not, continue to experience symptoms long after the infectious period has ended. These long-term effects are called post COVID-19 condition (PCC, or long COVID). The most common symptoms include fatigue, aches and pains in muscles or joints, breathlessness, headaches, and difficulty in thinking or concentrating. Post COVID-19 condition can affect a person's ability to perform daily activities such as work or household tasks.

Transmission

SARS-CoV-2, the virus causing COVID-19, spreads through the air via infectious respiratory particles released when an infected person breathes, talks, coughs, or sneezes. People are more likely to catch the virus when they are close to the infected person, or when they are sharing a closed space with them. The virus can also enter the body when someone touches contaminated surfaces like doorknobs or handles and then touches the eyes, nose or mouth.

SARS-CoV-2 circulates throughout the year, causing outbreaks worldwide without an established seasonal pattern.

Symptoms usually begin 3-6 days* after exposure (1) and last up to 10 days, with some individuals experiencing symptoms for a longer period. Infected people may also spread the virus in the absence of symptoms or in the days prior to developing any symptoms.

Treatment

Medical treatment exists for COVID-19, although most people recover from mild illness without needing treatment. For those who need treatment, health-care professionals suggest treatments for COVID-19 based on the severity of the disease and the risk of it getting worse. They consider the person's age, symptoms, any coexisting health conditions, and individual factors which might interfere with specific treatments.

Prevention

There are a number of things that people can do to prevent the spread of COVID-19. These are:

- if you are experiencing COVID-like symptoms or have tested positive for COVID-19, stay at home and away from others;
- avoid close contact with individuals who may be at high risk of developing complications if they get sick with COVID-19;
- if you must be around other people, wear a properly fitted mask if you experience any of the symptoms of COVID-19;
- cover your mouth and nose with a bent elbow or tissue when you cough or sneeze, and dispose of used tissues right away and clean your hands;
- clean your hands frequently with soap and water or alcohol-based hand rub; and
- Improve indoor ventilation, e.g. by opening windows and doors to increase natural airflow.

Those providing care for high-risk individuals should wear a medical mask and test for COVID-19 if they develop any of the symptoms of COVID-19.

There are a number of things that can be done to avoid becoming sick with COVID-19: These are:

- avoid close contact with people who are experiencing respiratory symptoms;
- avoid entering crowded or poorly ventilated areas if you are at high-risk of developing severe illness should you get sick with COVID-19; if you cannot avoid entering crowded or poorly ventilated areas, wear a properly fitted mask; and
- clean your hands frequently with soap and water or alcohol-based hand rub.

To prevent severe illness, hospitalization and death due to COVID-19, vaccination should be considered, particularly if you are at higher risk of serious illness, or if you are a health and care worker caring for those at higher risk.

Additional protective equipment should be considered for health workers taking part in aerosol-generating procedures, in consultation with occupational health and safety advisors.

Vaccines

Since their introduction, COVID-19 vaccines have saved millions of lives around the world by providing protection against severe disease, hospitalization, and death.

In November 2023, given that most of the global population had by that time been infected at least once, WHO updated its recommendations stating that a single dose of vaccine can be considered for those who have not yet been vaccinated. Revaccination against COVID-19 may be needed 6-12 months after the most recent dose for high priority groups such as older adults; individuals with severe obesity or multiple significant comorbidities, including

diabetes, chronic lung diseases, heart, liver and kidney diseases; persons with immunocompromising conditions; pregnant women and health and care workers with direct patient contact.

Even though vaccines protect against severe disease and death, they have limited impact on reducing transmission. It is still possible to spread COVID-19 to others if you are infected after being vaccinated.

WHO advises on COVID-19 vaccination through its Strategic Advisory Group of Experts on Immunization (SAGE). It is important to stay up to date with local recommendations provided by national health authorities.

WHO response

During the emergency phase of the COVID-19 pandemic, WHO coordinated the global response, supporting countries with preparedness, surveillance, diagnostics, clinical care, medical countermeasures, access to oxygen, country readiness and deployment of COVID-19 vaccines, vaccine safety monitoring, and evidence-based guidance. WHO recommended 13 vaccines for emergency use, guiding countries and UN agencies in their procurement, and led global coordination on research and development, and setting standards for vaccines, tests, and treatments. WHO also provided countries with strategic and policy recommendations, technical guidance, and tools.

The Access to COVID-19 Tools (ACT) Accelerator, launched by WHO and partners, was an unprecedented global effort to rapidly develop and deploy diagnostics, treatments, and vaccines. COVAX, its vaccine pillar, proved a ground-breaking global initiative to accelerate vaccine development and production, and ensure equitable global access to COVID-19 vaccines.

Since the Director-General of WHO lifted the Public Health Emergency of International Concern in May 2023 and the [emergency phase ended](#), WHO has continued to lead the global COVID-19 response, working with governments and partners to shift from emergency measures to long-term, integrated management of COVID-19 and other coronavirus threats. Ongoing activities in this respect include regular issuance of Disease Outbreak News posts, supporting ongoing research, updating guidance, monitoring virus trends, and evaluating vaccine composition.

To guide this shift, WHO has developed a [strategic and operational plan](#) to help countries manage COVID-19 as part of routine health systems. The plan promotes sustainable, evidence-based management of coronavirus threats, aligning COVID-19 responses with broader strategies for respiratory disease control and public health resilience.

*The Omicron variant virus strains, which now account for the vast majority of virus strains circulating in humans, have a shorter incubation period than pre-Omicron variant virus strains.
